COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	7/9/2026	**DATE SUBMITTED:**	7/30/2026
NAME:	Quioyo, Angelo M.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 1 — **Configure Network using Virtual Machines**

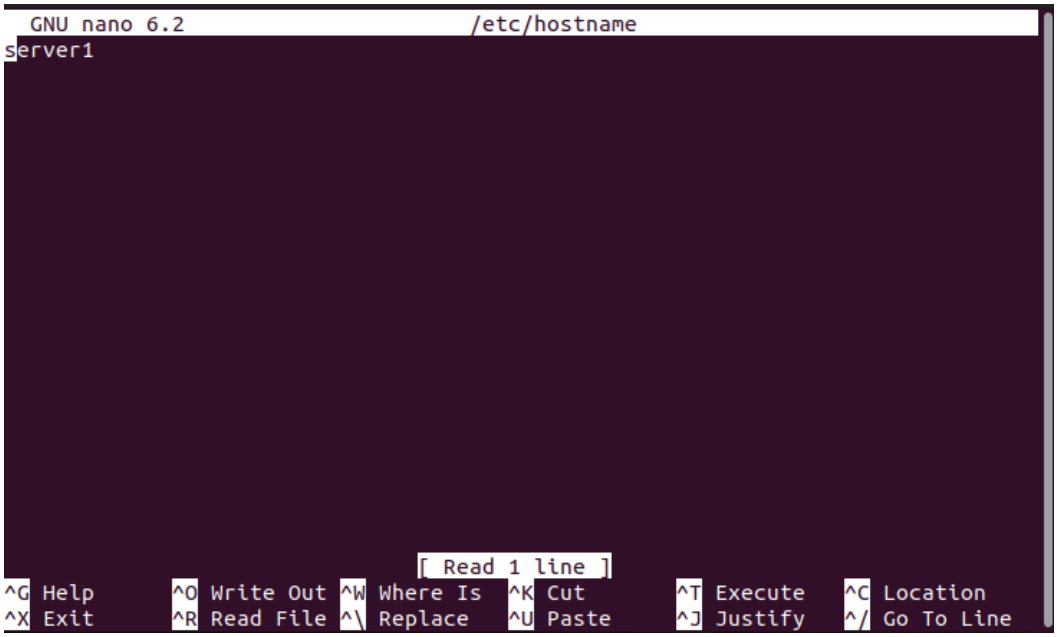
1. PROCEDURE OUTPUT

What procedures did you perform?

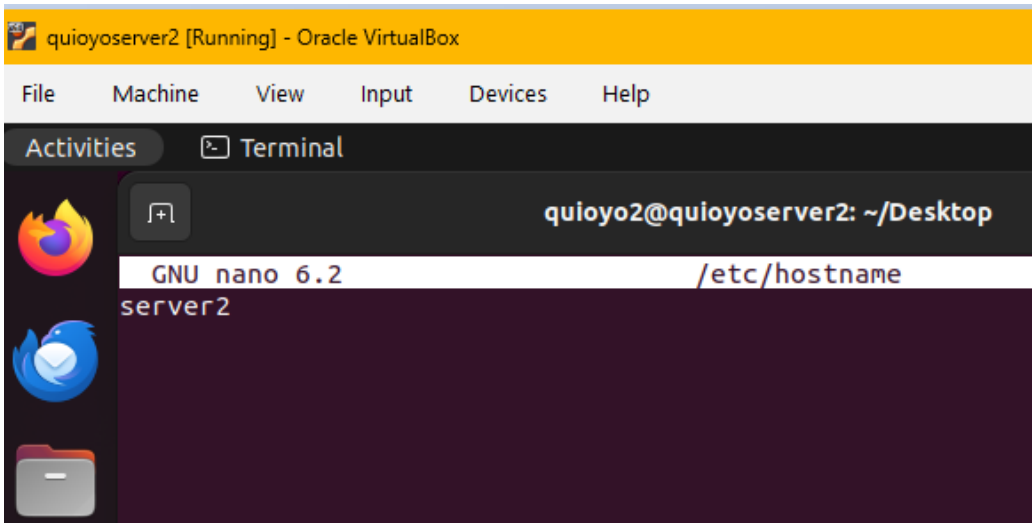
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Task 1: Do the following on Server 1, Server 2, and Local Machine. In editing the file using nano command, press control + O to write out (save the file). Press enter when asked for the name of the file. Press control + X to end.

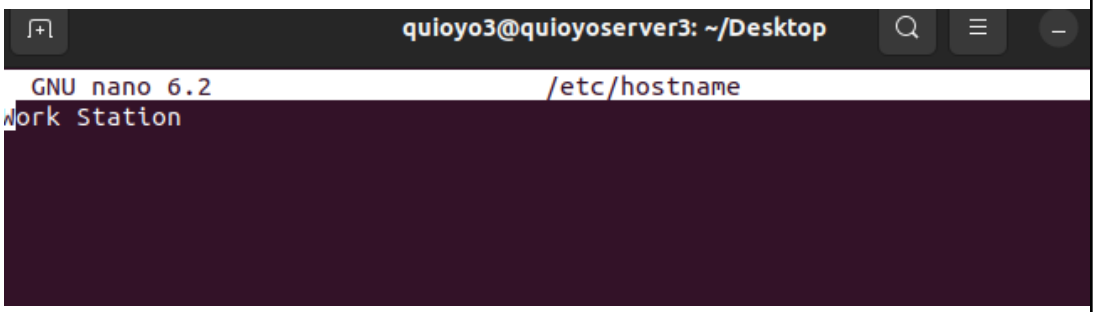
1. Change the hostname using the command *sudo nano /etc/hostname*
   1.1 Use server1 for Server 1



   1.2 Use server2 for Server 2

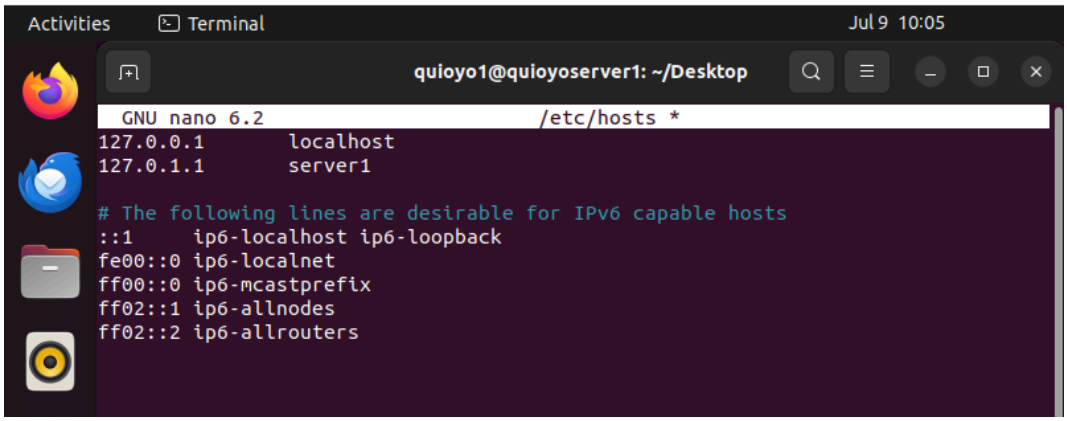


1.3 Use workstation for the Local Machine

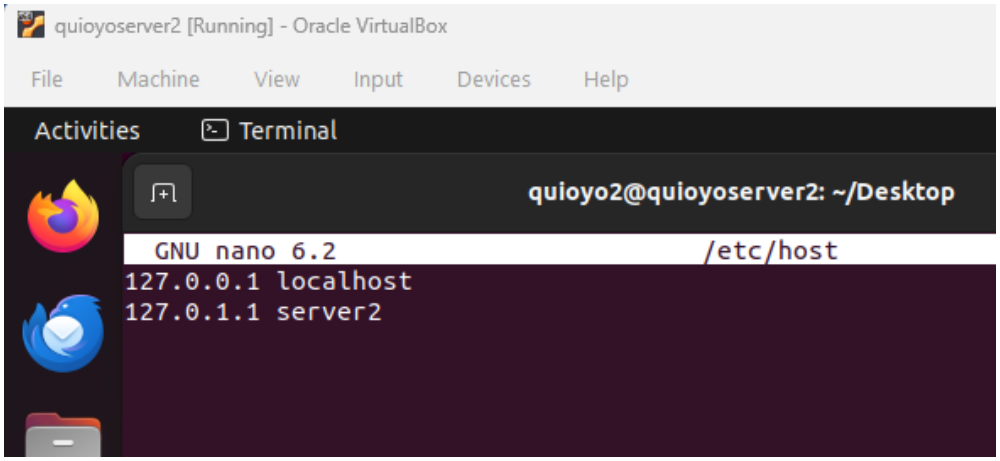


2. Edit the hosts using the command *sudo nano /etc/hosts*. Edit the second line.

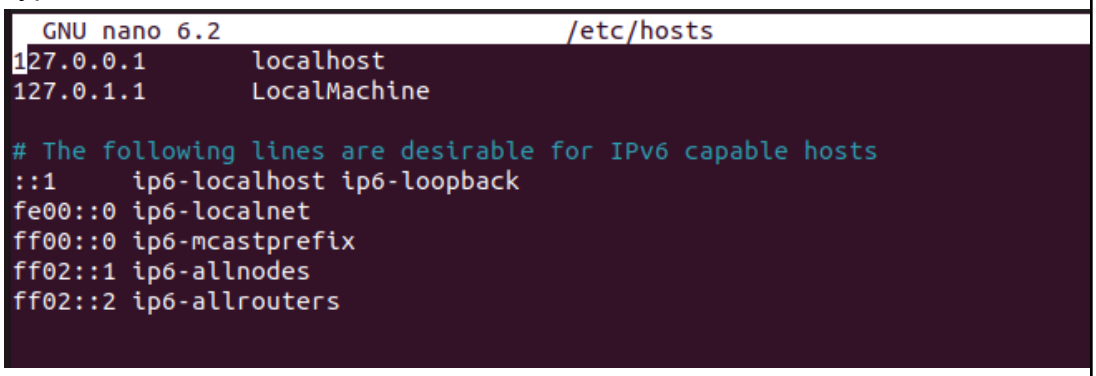
2.1 Type 127.0.0.1 server 1 for Server 1



2.2 Type 127.0.0.1 server 2 for Server 2



2.3 Type 127.0.0.1 workstation for the Local Machine



Task 2: Configure SSH on Server 1, Server 2, and Local Machine. Do the following:

1. Upgrade the packages by issuing the command *sudo apt update* and *sudo apt upgrade* respectively.

SERVER1:

```
quioyo1@quioyoserver1:~/Desktop$ sudo apt update && sudo apt upgrade -y
[sudo] password for quioyo1:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer requir
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

SERVER2:

```
quioyo2@quioyoserver2:~/Desktop$ sudo apt update && sudo apt upgrade -y
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
  linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic
  linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

LOCALMACHINE:

```
quioyo3@quioyoserver3:~/Desktop$ sudo apt update && sudo apt upgrade -y
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

2. Install the SSH server using the command *sudo apt install openssh-server*.

SERVER1:

```
quioyo1@quioyoserver1:~/Desktop$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-
server amd64 1:8.9p1-3ubuntu0.15 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-serve
r amd64 1:8.9p1-3ubuntu0.15 [435 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term
all 6.3-2ubuntu0.2 [267 kB]
```

SERVER2:

```
quioyo2@quioyoserver2:~/Desktop$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer requir
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
  linux-headers-5.19.0-32-generic linux-hwe-5.19-headers-5.19.0-32
  linux-image-5.19.0-32-generic linux-modules-5.19.0-32-generic
  linux-modules-extra-5.19.0-32-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-s
server amd64 1:8.9p1-3ubuntu0.16 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-s
```

LOCALMACHINE:

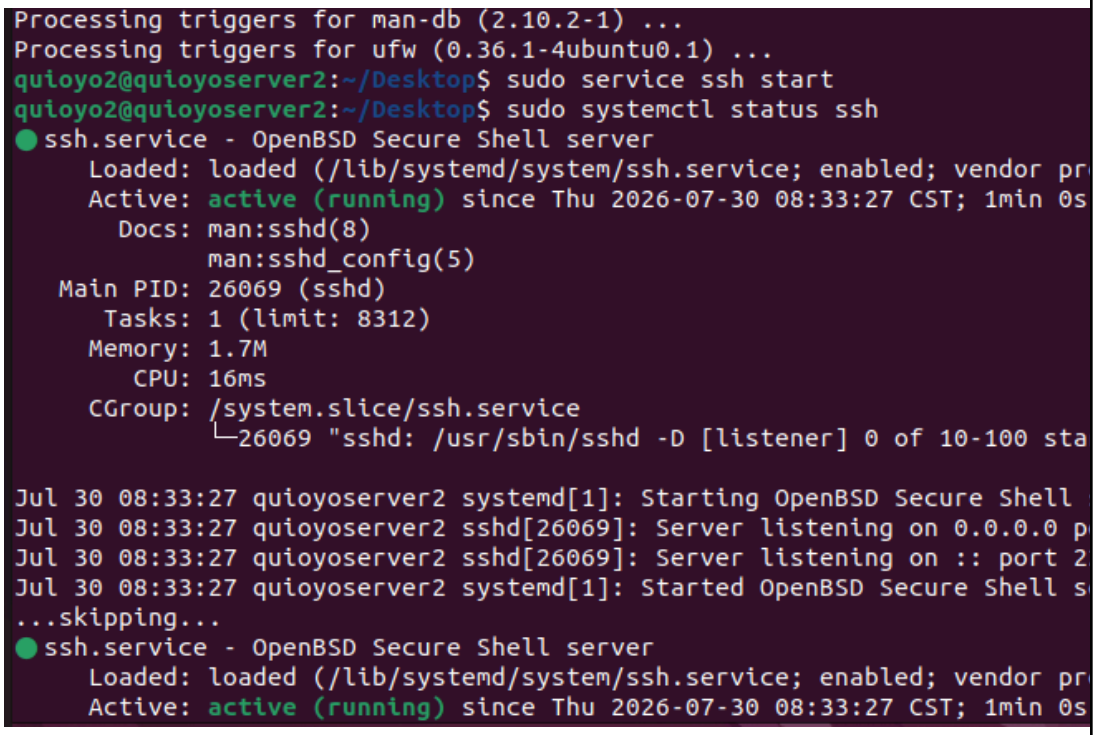
```
quioyo3@quioyoserver3:~/Desktop$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer requir
  libfwupd2 libfwupdplugin5 libgcab-1.0-0 libsmbios-c2
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 752 kB of archives.
After this operation, 6,051 kB of additional disk space will be used.
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-s
server amd64 1:8.9p1-3ubuntu0.15 [38.9 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-s
r amd64 1:8.9p1-3ubuntu0.15 [435 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-t
all 6.3-2ubuntu0.2 [267 kB]
```

3. Verify if the SSH service has started by issuing the following commands:
   3.1 sudo service ssh start
   3.2 sudo systemctl status ssh

SERVER1:



SERVER2:



LOCALMACHINE:

```
quioyo3@quioyoserver3:~/Desktop$ sudo service ssh start
quioyo3@quioyoserver3:~/Desktop$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: e>
     Active: active (running) since Thu 2026-07-09 10:10:11 CST; 4min 4s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 2776 (sshd)
      Tasks: 1 (limit: 8121)
     Memory: 1.7M
        CPU: 15ms
     CGroup: /system.slice/ssh.service
             └─2776 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 09 10:10:11 quioyoserver3 systemd[1]: Starting OpenBSD Secure Shell server.>
Jul 09 10:10:11 quioyoserver3 sshd[2776]: Server listening on 0.0.0.0 port 22.
Jul 09 10:10:11 quioyoserver3 sshd[2776]: Server listening on :: port 22.
Jul 09 10:10:11 quioyoserver3 systemd[1]: Started OpenBSD Secure Shell server.
quioyo3@quioyoserver3:~/Desktop$
```

4. Configure the firewall to all port 22 by issuing the following commands:
   4.1 sudo ufw allow ssh
   4.2 sudo ufw enable
   4.3 sudo ufw status

SERVER1:

```
quioyo1@quioyoserver1:~/Desktop$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
quioyo1@quioyoserver1:~/Desktop$ sudo ufw enable
Firewall is active and enabled on system startup
quioyo1@quioyoserver1:~/Desktop$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

SERVER2:

```
quioyo2@quioyoserver2:~/Desktop$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
quioyo2@quioyoserver2:~/Desktop$ sudo ufw enable
Firewall is active and enabled on system startup
quioyo2@quioyoserver2:~/Desktop$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)

quioyo2@quioyoserver2:~/Desktop$
```

LOCALMACHINE:

```
quioyo3@quioyoserver3:~/Desktop$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
quioyo3@quioyoserver3:~/Desktop$ sudo ufw enable
Firewall is active and enabled on system startup
quioyo3@quioyoserver3:~/Desktop$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
```

Task 3: Verify network settings on Server 1, Server 2, and Local Machine. On each device, do the following:

1. Record the ip address of Server 1, Server 2, and Local Machine. Issue the command *ifconfig* and check network settings. Note that the ip addresses of all the machines are in this network 192.168.56.XX.
   1.1 Server 1 IP address: 192.168.56.**109**
   1.2 Server 2 IP address: 192.168.56.**110**
   1.3 Server 3 IP address: 192.168.56.**108**
2. Make sure that they can ping each other.
   2.1 Connectivity test for Local Machine 1 to Server 1: ☐ Successful ☐ Not Successful

```
quioyo3@WorkStation:~/Desktop$ ping -c 4 192.168.56.109
PING 192.168.56.109 (192.168.56.109) 56(84) bytes of data.
64 bytes from 192.168.56.109: icmp_seq=1 ttl=64 time=5.18 ms
64 bytes from 192.168.56.109: icmp_seq=2 ttl=64 time=1.11 ms
64 bytes from 192.168.56.109: icmp_seq=3 ttl=64 time=1.90 ms
64 bytes from 192.168.56.109: icmp_seq=4 ttl=64 time=3.05 ms

--- 192.168.56.109 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3549ms
rtt min/avg/max/mdev = 1.105/2.809/5.180/1.533 ms
```

2.2 Connectivity test for Local Machine 1 to Server 2: ☐ Successful ☐ Not Successful

```
quioyo3@WorkStation:~/Desktop$ ping -c 4 192.168.56.110
PING 192.168.56.110 (192.168.56.110) 56(84) bytes of data.
64 bytes from 192.168.56.110: icmp_seq=1 ttl=64 time=2.15 ms
64 bytes from 192.168.56.110: icmp_seq=2 ttl=64 time=1.01 ms
64 bytes from 192.168.56.110: icmp_seq=3 ttl=64 time=1.52 ms
64 bytes from 192.168.56.110: icmp_seq=4 ttl=64 time=0.917 ms

--- 192.168.56.110 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4235ms
rtt min/avg/max/mdev = 0.917/1.396/2.148/0.490 ms
```

2.3 Connectivity test for Server 1 to Server 2: ☐ Successful ☐ Not Successful

```
quioyo1@server1:~/Desktop$ ping -c 4 192.168.56.110
PING 192.168.56.110 (192.168.56.110) 56(84) bytes of data.
64 bytes from 192.168.56.110: icmp_seq=1 ttl=64 time=1.54 ms
64 bytes from 192.168.56.110: icmp_seq=2 ttl=64 time=0.912 ms
64 bytes from 192.168.56.110: icmp_seq=3 ttl=64 time=1.70 ms
64 bytes from 192.168.56.110: icmp_seq=4 ttl=64 time=1.37 ms

--- 192.168.56.110 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 0.912/1.381/1.698/0.294 ms
```

Task 4: Verify SSH connectivity on Server 1, Server 2, and Local Machine.

1. On the Local Machine, issue the following commands:

1.1 ssh username@ip_address_server1 for example, *ssh jvtaylar@192.168.56.120*

1.2 Enter the password for server 1 when prompted

1.3 Verify that you are in server 1. The user should be in this format user@server1. For example, *jvtaylar@server1*

2. Logout of Server 1 by issuing the command *control + D.*

```
quioyo3@WorkStation:~/Desktop$ ssh quioyo1@192.168.56.109
quioyo1@192.168.56.109's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-124-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

116 updates can be applied immediately.
109 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

quioyo1@server1:~$
logout
Connection to 192.168.56.109 closed.
```

3. Do the same for Server 2.

```
quioyo3@WorkStation:~/Desktop$ ssh quioyo2@192.168.56.110
The authenticity of host '192.168.56.110 (192.168.56.110)' can't be established.
ED25519 key fingerprint is SHA256:JhsbTLfniSBzDjBro/r5Up2DFiKwaSWbwQeJPQFwXWc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.110' (ED25519) to the list of known hosts
quioyo2@192.168.56.110's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


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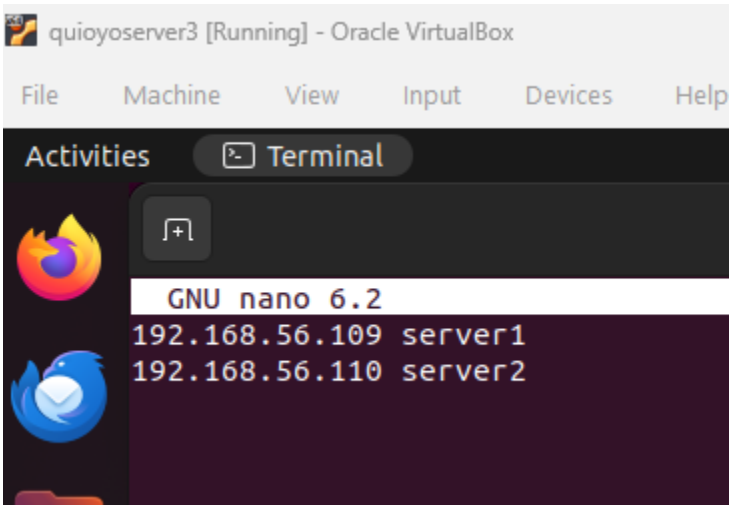
quioyo2@server2:~$
logout
Connection to 192.168.56.110 closed.
```

4. Edit the hosts of the Local Machine by issuing the command *sudo nano /etc/hosts.* Below all texts type the following:

4.1 IP_address server 1 (provide the ip address of server 1 followed by the hostname)

4.2 IP_address server 2 (provide the ip address of server 2 followed by the hostname)

4.3 Save the file and exit.



5. On the local machine, verify that you can do the SSH command but this time, use the hostname instead of typing the IP address of the servers. For example, try to do *ssh jvtaylar@server1*. Enter the password when prompted. Verify that you have entered Server 1. Do the same for Server 2.

Reflections:

Answer the following:

1. How are we able to use the hostname instead of IP address in SSH commands?
   I am able to use the hostname instead of the IP addresses manually by using /etc/hosts because it acts as a local domain name system lookup table that can map names to a specific IP address before querying. By entering the ssh command with a hostname, the system checks the local mapping file to automatically resolve the name to its corresponding IP address.
2. How secured is SSH?

   It uses strong encryption protocols by establishing an encrypted tunnel that scrambles data and commands transmitted across the network.

COMPUTER ENGINEERING

QUEZON CITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Guidance of typing what commands to use correctly**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Command prompts are only followed and did not need more modification, I only need to follow what to type**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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